

Case 4 (Subgraph Matching)

G-Q: Determine if a smaller graph is present as an exact match within a larger graph. In a directed graph, $(i \rightarrow j)$ means that node i and node j are connected with a directed edge from node i to node j . Given a graph G and a subgraph G' , you need to output Yes or No, indicating whether subgraph G' is present within the directed graph G . Q: The nodes of graph G are numbered from 0 to 8, and the edges are: $(0 \rightarrow 1)$ $(0 \rightarrow 8)$ $(0 \rightarrow 5)$ $(0 \rightarrow 6)$ $(0 \rightarrow 4)$ $(1 \rightarrow 3)$ $(1 \rightarrow 5)$ $(1 \rightarrow 6)$ $(2 \rightarrow 6)$ $(2 \rightarrow 3)$ $(3 \rightarrow 6)$ $(3 \rightarrow 5)$ $(3 \rightarrow 4)$ $(5 \rightarrow 7)$ $(7 \rightarrow 8)$. The nodes of subgraph G' are numbered from a to e , and the edges are: $(a \rightarrow e)$ $(a \rightarrow d)$ $(b \rightarrow c)$ $(c \rightarrow e)$. **Is subgraph G' present within graph G as a direct substructure?**

GraphWiz: To find if subgraph G' is present in graph G , we need to identify a node in G that has two outgoing edges to different nodes, just like 'a' in G' . Then, we need to check if one of those nodes has another outgoing edge to a different node, just like 'e' in G' . Finally, we need to check if the other node has an incoming edge from a different node, just like 'd' in G' . Looking at graph G : Node 0 has five outgoing edges, but none of them lead to a node that has another outgoing edge to a different node, so node 0 cannot be 'a'. Node 1 has three outgoing edges, but none of them lead to a node that has another outgoing edge to a different node, so node 1 cannot be 'a'. Node 2 has two outgoing edges, to nodes 6 and 3. Node 6 has no outgoing edges, so it cannot be 'e'. Node 3 has three outgoing edges, to nodes 6, 5, and 4. Node 6 cannot be 'e', since it has no outgoing edges. Node 5 has one outgoing edge, to node 7, which could be 'e'. Node 4 has no outgoing edges, so it cannot be 'e' or 'd'. Therefore, node 2 cannot be 'a'. Node 3 has four outgoing edges, to nodes 6, 5, 4, and 7. Node 6 cannot be 'e', since it has no outgoing edges. Node 5 has one outgoing edge, to node 7, which could be 'e'. Node 4 has no outgoing edges, so it cannot be 'e' or 'd'. Node 7 has one outgoing edge, to node 8, which could be 'e'. Therefore, node 3 could be 'a', with nodes 5 and 7 being 'e' and 'd' respectively, or vice versa. Checking the connections of node 3's targets: Node 3 to 5 (potential 'e') Node 3 to 7 (potential 'd') Node 5 to 7 (matches 'e' to 'd') Node 7 to 8 (potential 'e') Node 5 to 8 (potential 'd') Node 8 has no outgoing edges, so it cannot be 'e'. Node 5 has one outgoing edge, to node 7, which could be 'e'. Therefore, node 7 cannot be 'd', since it has no incoming edges. Node 5 could be 'd', since it has an incoming edge from node 3. Thus, the sequence $3 \rightarrow 5 \rightarrow 7$ in G corresponds to the sequence $a \rightarrow e \rightarrow d$ in G' , and the sequence $3 \rightarrow 7 \rightarrow 8$ in G corresponds to the sequence $a \rightarrow d \rightarrow e$ in G' , which means subgraph G' is present as a direct substructure in graph G . ### Yes. 